Gaussian Process and Kalman Baselines for Bradyseismic Signal Extraction from GNSS Time Series at Campi Flegrei

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Abstract. Monitoring ground deformation at Campi Flegrei with GNSS requires separating the slow bradyseismic deformation signal from the trend, the seasonality, and the stochastic noise that overwhelm it. This work addresses this problem on an hourly dataset of seven GNSS stations in baseline configuration, three components each, spanning approximately four years of observations. We present a preprocessing chain that removes the deterministic components and the measurement artifacts and characterizes the residual noise as power-law plus white noise. We then construct two inference baselines for extracting the deformation signal: a Gaussian Process in batch formulation and a state-space Kalman filter, both grounded in the same generative model, so as to isolate the effect of the inference engine alone. The comparison, carried out on the residual spectrum, on the estimated noise parameters, and on the extracted signal, shows that the two baselines preserve the spectral slope to the third decimal place, agree on the power-law noise correction factor, and extract the same deformation signal, with sub-millimeter differences against a signal of tens-of-centimeters magnitude. This agreement holds despite the two methods differing in kernel, regularization, temporal resolution, and prior distribution, and indicates that, for these data, the extracted deformation signal is robust to the choice of estimation method.

Keywords: Gaussian Processes · Kalman Filter · GNSS Time Series · Bradyseism

1 Introduction

Campi Flegrei is a volcanic caldera of high dynamic complexity, characterized by multi-decadal cycles of ground uplift and subsidence known as bradyseism [7]. Continuous monitoring of deformation is essential for surveillance of the area , and GNSS (Global Navigation Satellite Systems) constitute one of its primary tools [12], providing ground displacement time series at high temporal resolution and across multiple components. Extracting the deformation signal of interest from these series is, however, made non-trivial by the simultaneous presence of several contributions: in addition to volcanic deformation, GNSS series contain a long-period trend and seasonal components, whose removal is necessary because periodic signals distort velocity estimation [3], measurement artifacts (outliers and discontinuities), and a stochastic noise component. The presence of outliers and anomalies, common in real-world time series, in turn requires dedicated detection procedures before any modeling [9, 4].

The latter is particularly insidious. It is well known that noise in GNSS series is not purely white but exhibits a correlated power-law structure (colored noise), whose mischaracterization leads to a systematic underestimation of the uncertainties on the estimated quantities, starting with the velocities [34, 21]. The noise structure is moreover not uniform: spatial variations among stations of the same network have been documented [15], as well as temporal variations of its properties [27]. Correct modeling of colored noise is therefore a prerequisite for any quantitative analysis of deformation [16], and dedicated tools such as Hector [5] allow its parameters to be estimated by maximum likelihood.

In this context, the problem of extracting the slow deformation signal can be formulated as a denoising problem: separating the smooth deformation component from the power-law and white noise that overwhelm it at high frequencies. Approaches based on Gaussian Processes have been successfully employed for the reconstruction of geodetic time series, even in the presence of missing data and time-varying seasonal signals [35], whose non-stationary nature has motivated the use of techniques such as singular spectrum analysis to extract their time-varying periodic component [31, 29], while spatial decomposition techniques such as PCA and ICA have been applied to the analysis of ground deformation [13, 6]. The choice of estimation method, however, introduces a methodological degree of freedom: for a given signal and noise model, different inference formulations could in principle produce different estimates, and it is not obvious a priori how much the extracted signal depends on that choice.

This work addresses this question on a GNSS dataset of Campi Flegrei. The contributions are threefold. First, we present a complete preprocessing chain that removes the deterministic components and the measurement artifacts and characterizes the residual noise. Second, we construct two inference baselines for extracting the deformation signal, namely a Gaussian Process in batch formulation and a state-space Kalman filter, deliberately grounded in the same generative model, so as to make the comparison controlled by construction. Third, we show empirically that, for these data, the two baselines produce substantially the same extracted signal and the same correction of the noise parameters, de-

spite differing in inference engine, kernel, regularization, and temporal resolution: the extracted deformation signal is therefore robust to the choice of estimation method.

The remainder of the article is organized as follows. Section 2 describes the dataset and the preprocessing chain, including the characterization and spectral analysis of the noise. Section 3 formulates the two inference baselines and discusses their comparability. Section 4 presents the results of the comparison. Section 5 discusses their implications and Section 6 concludes by indicating future developments.

2 Data and preprocessing

The analysis is based on the displacement time series of seven permanent GNSS stations in the Campi Flegrei area, provided by the Istituto Nazionale di Geofisica e Vulcanologia Osservatorio Vesuviano (INGV-OV). Before the slow deformation signal can be extracted (Section 3), the series go through a preprocessing chain that removes the deterministic components and the measurement artifacts and that characterizes the residual stochastic noise. This section describes the dataset and each stage of the chain; the limitations of the adopted choices are collected in Section 2.7.

2.1 Dataset

We consider the seven stations ACAE, RITE, STRZ, PIS1, IPPO, VICA and ARFE, each analyzed in baseline with respect to the reference station LICO. For each station the three displacement components are available, the horizontal East and North and the vertical (Up), for a total of $7 \times 3 = 21$ series. The data are sampled at hourly cadence and cover the interval from April 2022 to January 2026, about three years and nine months of continuous observations. The series contain some epochs with missing data, distributed non-uniformly, handled in the subsequent stages of the pipeline. Each series is preliminarily referred to a common origin by subtracting its own initial level, so as to work on relative displacements.

2.2 Outlier removal

Outlier removal is performed in an unsupervised manner with the ADTK library [1], applied in three successive steps to each series. Since the detectors operate on assumptions of local stationarity, in each step the missing values are masked and the series is temporarily deprived of its own trend: an unremoved background drift would alter the estimation of the thresholds and produce false positives. The first step employs a threshold detector on global quantiles (`QuantileRangeAD`), necessary because some stations such as STRZ–LICO (Figure 1c) present isolated spikes of very large amplitude that, if not eliminated first, would dominate the statistics on which the subsequent detectors are based.

In the stations free of such extreme spikes this step produces no relevant effects, since any moderate outliers would in any case be captured by the interquartile detector applied subsequently: its function is therefore specifically to neutralize the anomalous peaks that would otherwise distort the downstream thresholds. The second step concerns specifically the station ACAE, where the seismic swarm of September 2025 produced a level shift (offset) of an amplitude such as to compromise the estimation of outliers over the entire series; the offset was removed manually (see Section 2.7 for the implications of this choice). The third step applies a detector based on the interquartile range (`InterQuartileRangeAD`), which identifies the residual outliers of moderate amplitude. The effect of the procedure on the series is illustrated in Figure 1, which compares the three components of the stations ACAE–LICO and STRZ–LICO before and after removal. At a qualitative level, the anomalous spikes present in the raw series are correctly removed: both the isolated peaks of extreme amplitude (STRZ–LICO) and the level shift of September 2025 (ACAE–LICO) disappear, while the underlying dynamics of the signal remains preserved. The effectiveness is moreover quantified by monitoring the evolution of the shape statistics of the residual distribution through the three steps: both the skewness and the kurtosis stabilize after the complete sequence, and the tails of the empirical probability density progressively shrink (Figure 2), a sign that the outliers have been removed without distorting the central body of the distribution.

2.3 Detrending

The long-period trend component is estimated and removed by means of STL decomposition (Seasonal-Trend decomposition based on Loess) [10]. The hourly series is first resampled at daily scale, on which STL estimates a slow and regular trend component with a wide window (of biennial order); the daily trend thus obtained is then mapped back onto the original hourly index and subtracted. On the components without a significant trend STL is not applied, but the median is simply removed, centering the series without imposing a behavior that the data do not support. The effect of detrending on the residual distribution is diagnostic. In the components characterized by a marked trend, the series show, before removal, a skewness close to zero but a kurtosis around -1: a platykurtic and effectively multimodal distribution, whose shape reflects the presence of the underlying trend, which disperses the values across several modes. After detrending the density collapses onto a single mode (Figure 3), confirming that the multimodal structure was attributable to the trend. The relation holds, however, in one direction only: a marked trend produces multimodality, but unimodality does not imply the absence of trend. Some components with a weak trend, such as IPPO–LICO height, present an already unimodal distribution upstream, because the trend is too small to separate the values into distinct modes; others, free of trend and centered with the median alone (such as ACAE–LICO north or VICA–LICO east), are unimodal from the start with a kurtosis close to 0.5. In both cases the removal does not substantially alter the shape of the density:

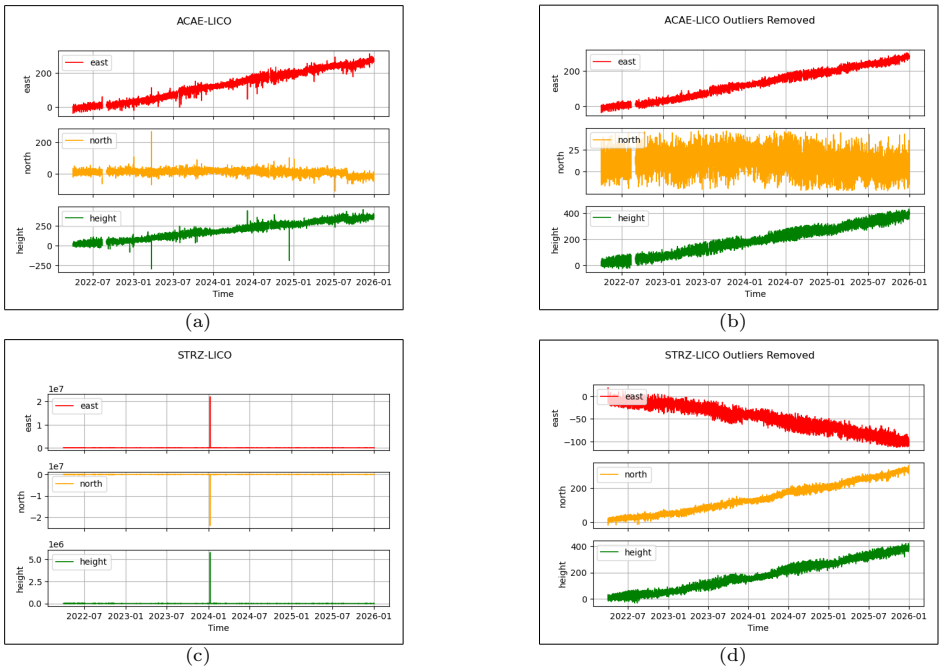


Fig. 1: Time series of the three components (East, North, Up) before and after outlier removal, for the stations ACAE–LICO (a, b) and STRZ–LICO (c, d). On the left the raw series, on the right the cleaned series.

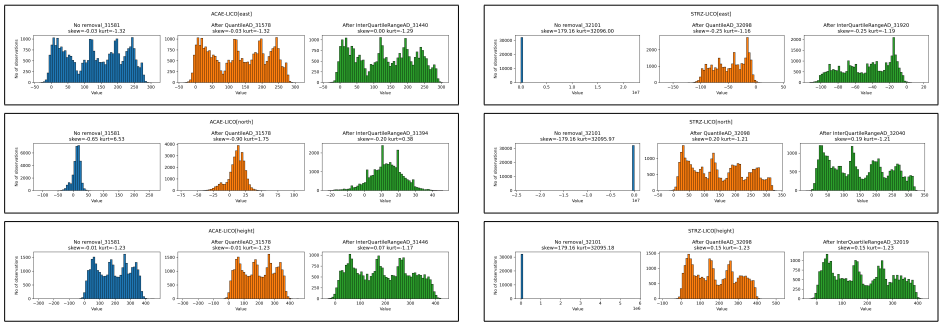


Fig. 2: Empirical distribution of the residuals through the three removal steps (no removal, after QuantileAD, after InterQuartileRangeAD) for the east, north and height components of the stations ACAE–LICO (left column) and STRZ–LICO (right column).

a behavior that is expected and consistent with the absence of a multimodal structure to remove.

2.4 Deseasonalization

The removal of seasonality is entrusted to an MSTL decomposition (Multiple Seasonal-Trend decomposition using Loess) [2], an extension of STL to the case of

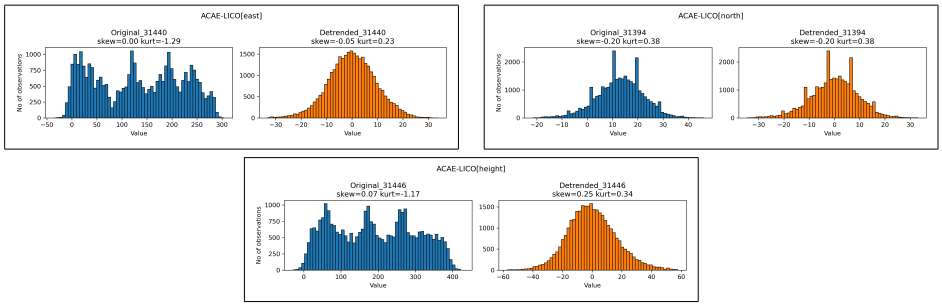


Fig. 3: Empirical distribution of the residuals before and after trend removal for the east, north and height components of the station ACAE–LICO.

several superimposed periodic components. The seasonal periods to be modeled are identified beforehand by computing a Lomb–Scargle periodogram [23, 30], suited to series that are potentially irregular owing to the missing values, and by identifying the peaks corresponding to the known periodicities (annual and semi-annual). These periods are passed to MSTL, which operates at weekly scale and estimates a seasonal component for each; the sum of the seasonal components is finally mapped back onto the hourly index and subtracted. The use of MSTL allows the estimated seasonality to adapt over time, following the amplitude variations of the periodic signal rather than imposing a rigid shape.

2.5 Noise characterization

The residual stochastic noise is characterized with Hector [5], which estimates its model by maximum likelihood. A Power-law plus White model common to all three components is adopted, which returns for each series the spectral index κ, the amplitude σ_{pl} of the power-law component and the amplitude σ_w of the white noise. These three parameters constitute the common initialization of the two signal estimators described in Section 3: they fix the spectral shape of the colored noise and the starting level of the white noise. The estimated spectral indices fall in an interval between $\kappa \approx -0.5$ (upper limit) and $\kappa \approx -0.9$, indicating a colored-noise regime consistent across the whole network but with a non-negligible variability between stations and components. The steepest values (κ close to -0.9) correspond to the situations in which the power-law component tends to be overestimated, as discussed among the limitations of the method.

2.6 Spectral analysis

The spectral analysis of the residuals closes the preprocessing and provides two pieces of information that guide the subsequent modeling: the location in frequency of the slow signal and the verification of the noise model. The spectrograms (Figure 4, left column) show that the low-frequency power concentrates at periods longer than a month, with the most intense contributions around 10^{-2}

cycles/day (about one hundred days). The slow deformation component of interest therefore lives on time scales of the order of a few months (typical lengthscale between one and six months), clearly separated from the high-frequency band dominated by the noise. This scale separation justifies the subsampling adopted in the training of the Gaussian Process (Section 3.2): the subsampling step remains well below the time scale of the signal, whose frequency is far lower than the induced Nyquist frequency, so that the information useful for the reconstruction of $f(t)$ is not removed. The power spectral density (Figure 4, right column) allows the empirical slope of the residuals to be compared with the power-law index estimated by Hector. The agreement is close on the East component (slope -0.504 against $\kappa = -0.536$), whereas on the North and vertical components the Hector index turns out to be appreciably steeper than the measured slope (-0.675 against $\kappa = -0.778$ for North; -0.757 against $\kappa = -0.795$ for the vertical). This discrepancy constitutes the first indication of an overestimation of the power-law component by Hector, the more marked the steeper the estimated index. It must moreover be emphasized that any agreement between the empirical slope and the index κ would in any case concern only the spectral shape of the colored noise and would not constitute a guarantee of the correctness of the estimation model: in particular it does not constrain the partition of power between the power-law component and the white noise. The phenomenon and its consequences for that partition are taken up again among the limitations of the preprocessing (Section 2.7) and quantified on the results (Section 4).

2.7 Limitations of the preprocessing

The chain described is conceived as a solid but deliberately simple baseline; some of its choices constitute known limitations, which are reported for transparency and which guide future developments.

- **Manual offset correction.** The offset induced by the seismic swarm on ACAE was removed manually. The solution is adequate as a first step but not systematic; a planned development is an automatic detection of the offsets, facilitated by the resampling of the series from hourly to daily (the scale at which the offsets are more clearly visible), with removal and documented tracking of each jump.
- **Empirical rather than tectonic trend.** The removed trend is estimated in a data-driven manner by STL and does not correspond to a physically determined tectonic velocity. A correction based on a tectonic trend provided by INGV, currently unavailable, would be preferable; alternative approaches (for example a PCA) for the estimation of the non-tectonic trend component should moreover be explored.
- **Seasonality: identification and modeling.** The identification of the periods is based on Lomb–Scargle, which assumes periodicities of constant amplitude and frequency over time. At Campi Flegrei the bradyseismic modulation violates this assumption: the power of a cycle at variable frequency does not concentrate in a sharp peak but is distributed over a band, with the

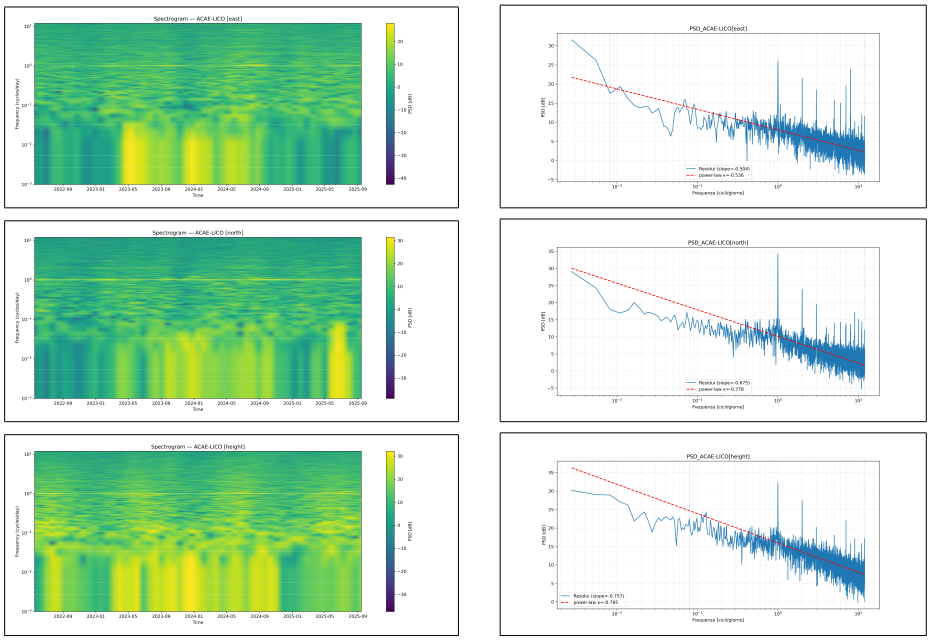


Fig. 4: For the station ACAE–LICO, spectrogram (left) and power spectral density of the residuals (right) for the east, north and height components.

maximum possibly turning out broadened and shifted with respect to the nominal period. Furthermore, at periods comparable with the duration of the series (beyond ∼457 days, corresponding to the criterion of 3 cycles over ∼4 years of data) the periodogram no longer discriminates a real periodicity from the low-frequency tail of the colored noise or from a residual trend: the maxima that emerge in that region are likely artifacts, and treating them as seasonality would mean modeling noise. For this reason the candidate periods are bounded from above and the measured period is passed to MSTL instead of the canonical one. The removal is then entrusted to MSTL, which captures the seasonality flexibly with a time-varying shape, mitigating but not entirely eliminating these effects; source-separation approaches (for example FastICA [18]) could jointly extract periodic components and nonlinear trend, and this is a direction of development.

– **Noise model.** The choice of a Power-law plus White model common to all components is not necessarily optimal. There is evidence that Hector tends to overestimate the power-law component, in particular where the estimated spectral index is steeper (κ close to -0.9): the estimators of Section 3, and in particular the Gaussian Process, redistribute power from the power-law to the white noise, whose estimated amplitude grows appreciably with respect to the Hector value. This is the reason for the constraints (floor on the white noise) introduced in the estimation. Alternative noise models remain to be explored.

3 Extraction of the non-deterministic signal: two inference baselines

After the removal of the deterministic components described in Section 2 (trend and seasonality), each residual series $y(t)$ is modeled as the superposition of three contributions:

$$y(t) \;=\; f(t) \;+\; n_{\mathrm{pl}}(t) \;+\; n_w(t), \tag{1}$$

where $f(t)$ is the *slow deformation signal*, that is, the residual bradyseismic modulation of geophysical interest, $n_{\mathrm{pl}}(t)$ is a power-law *colored noise* whose spectral index and global amplitude are estimated upstream by means of Hector [5] (cf. Sec. 2), and $n_w(t)$ is a Gaussian *white noise* of constant variance. The goal of the extraction is to isolate $f(t)$ from the two stochastic terms.

We address the problem with two distinct estimators: a Gaussian Process (GP) [25] in batch formulation and a Kalman filter [19] in recursive state-space formulation. The choice of placing two methods side by side does not aim to establish a superiority of one over the other, but rather to verify how much the estimate of $f(t)$ depends on the adopted inference engine, for the same generative model and the same noise characterization. For the Campi Flegrei GNSS data considered here, the agreement of the two estimators (Section 4) supports the robustness of the extracted signal with respect to this methodological choice.

3.1 Shared generative model

Both baselines assume that the deformation signal $f(t)$ is a smooth function of time. This assumption is consistent with the nature of bradyseism, which manifests as a regular deformation free of discontinuities once the instrumental and co-seismic offsets have been corrected in the preprocessing stage. The colored component $n_{\mathrm{pl}}(t)$ is described as the sum of N independent Ornstein–Uhlenbeck (OU) processes [20],

$$n_{\mathrm{pl}}(t) \;=\; \sum_{i=1}^{N} u_i(t), \qquad \mathrm{cov}[u_i(t), u_i(t')] \;=\; w_i\, e^{-a_i\,|t-t'|}, \tag{2}$$

whose rates a_i and weights w_i are fixed upstream so as to approximate the f^{κ} spectrum estimated by Hector over a chosen frequency band. The sum (2) constitutes a finite-band approximation of the power-law noise: its spectral *shape* is imposed by κ, while its overall amplitude is governed by a single global scale factor, left free during estimation to correct a possible overestimation of the amplitude by Hector. The white term $n_w(t)$ is parametrized by a single variance, constrained not to fall below a physical floor that prevents its collapse to zero.

The parameters of the colored component (spectral index, rates and weights of the OU processes) and the white-noise floor are inherited from the same MLE characterization by Hector and kept *identical* for the two methods. The two baselines therefore share the generative model (1), the noise parametrization (2), and the input data; what is described in Sections 3.2–3.4 concerns exclusively the inference engine with which $f(t)$ is reconstructed.

3.2 Gaussian Process baseline

The first baseline addresses the problem in batch form [25]. The signal $f(t)$ and the two noise terms are modeled jointly through an additive covariance function,

$$k(t,t') \;=\; \underbrace{k_{\mathrm{RBF}}(t,t')}_{\text{signal}} \;+\; \underbrace{s\sum_{i=1}^{N} w_i\, e^{-a_i|t-t'|}}_{\text{power-law}} \;+\; \underbrace{\sigma_w^2\,\delta_{tt'}}_{\text{white}}, \tag{3}$$

in which the RBF (radial basis function) kernel describes the slow component, the sum of exponentials realizes the power-law (2) with global scale factor s, and the diagonal term represents the white noise. The RBF lengthscale is constrained within an interval $[\ell_{\min}, \ell_{\max}]$ chosen so as to reserve the low frequencies for the signal and to prevent it from competing with the high-frequency tail of the power-law. The estimate of $f(t)$ is the posterior mean of the process conditioned on the entire series. Denoting by $\mathbf{K}_{\mathrm{tot}}$ the full covariance matrix evaluated at the training points and by $\mathbf{K}_s$ that of the signal component alone, we have

$$\widehat{f} \;=\; \mathbf{K}_s\left(\mathbf{K}_{\mathrm{tot}} + \sigma_w^2\mathbf{I}\right)^{-1}\mathbf{y}, \tag{4}$$

with uncertainty obtained from the corresponding posterior covariance. The inference is *global*: each epoch contributes to the estimate of all the others through the inversion of $\mathbf{K}_{\mathrm{tot}}+\sigma_w^2\mathbf{I}$. The free hyperparameters, that is, lengthscale, signal amplitude, power-law scale factor s, and white-noise variance σ_w^2, are estimated by maximizing the marginal log-likelihood. On two of them a weak prior regularization acts, realized with LogNormal distributions: one on the lengthscale, centered on the initialization value, and one on s, centered on unity, so as to favor the power-law amplitude estimated by Hector in the absence of contrary evidence in the data. The objective actually maximized is therefore the penalized log-likelihood $\log p(y|\theta) + \log p(\theta)$, corresponding to a MAP (maximum a posteriori) estimate rather than a pure maximum-likelihood one. Since the cost of the inversion in (4) scales as $\mathcal{O}(n^3)$ in the number of epochs, the GP is trained on a regular subsampling of the series and subsequently evaluated on the entire time axis. This subsampling does not compromise the signal estimate: the spectral analysis of Section 2 places the slow component of interest at periods longer than the daily window, well beyond the Nyquist frequency induced by the subsampling step, which therefore does not remove information useful for the reconstruction of $f(t)$.

3.3 Kalman baseline

The second baseline reformulates the same additive model (1) in state space and performs the inference recursively [19, 28]. For the slow signal a Matérn-5/2 kernel is adopted: unlike the RBF, which as an infinitely smooth limit does not admit an exact finite state-space representation, the Matérn-5/2 admits an exact finite-dimensional state-space representation [14] (3 components: value, velocity,

and acceleration of the signal) and closely approximates its regularity, the RBF being the $\nu \to \infty$ limit of the Matérn family [25]. The exact discretization of the transition and process-noise matrices is obtained by matrix means (matrix exponential and Van Loan formula [33]), so as to preserve the equivalence with the continuous-time formulation at any sampling step. The power-law component is represented by the *same* OU processes (2) as in the GP baseline, each cast as a scalar state with transition $e^{-a_i\,\Delta t}$ and process noise $w_i\,(1 - e^{-2a_i\,\Delta t})$. The augmented state concatenates the three Matérn-5/2 components and the N OU states; the observation matrix sums the signal value and the contributions of all the OU processes, exactly reproducing the measurement model (1). The white noise $n_w(t)$ enters as measurement variance R.

The estimate of $f(t)$ is obtained with a forward pass of the filter, which provides the log-likelihood for the estimation of the parameters, followed by an RTS smoother (Rauch–Tung–Striebel [26]) that propagates the information backward and returns the smoothed state with its uncertainty. The signal component is the first coordinate of the smoothed state. Unlike the GP, the cost of the recursive inference scales linearly in the number of epochs, and the filter therefore operates at full temporal resolution on the entire series. The free parameters (lengthscale and amplitude of the signal, power-law scale factor, white-noise variance) are estimated by maximum likelihood subject to box constraints only: the lengthscale within $[\ell_{\min}, \ell_{\max}]$, the scale factor within a positive interval, the white noise above the floor, without any prior penalization term.

3.4 Comparability by construction

The two baselines share the generative model (1), the noise parameters estimated by Hector, the admissible support of the parameters, and the input data. They differ in the inference engine, global and batch in the GP, recursive and state-space in the Kalman, and in three choices dictated by it:

1. the **signal kernel**: RBF in the GP, Matérn-5/2 in the Kalman, a difference imposed by the requirement of state-space representability and not by a different assumption on the signal [28]. It follows that the smoothness hyperparameters of the two kernels (lengthscale and amplitude) are not directly comparable between the two methods, as they parametrize different covariance functions: the equivalence between the baselines must therefore be verified on the extracted signal $f(t)$ and on the noise parameters, not on the individual hyperparameters of the signal kernel;
2. the **regularization**: weak LogNormal priors on the lengthscale and on the power-law scale factor in the GP (MAP estimate), against box constraints only in the Kalman (pure maximum likelihood). The GP thus adds a soft push toward preferred prior values within the admissible interval, whereas the Kalman lets the data alone decide within the same limits;
3. the **temporal resolution**: the GP is trained on a subsampling owing to the cubic cost and the amount of VRAM, the Kalman operates at full resolution.

These are therefore two formulations that share the framing of the problem but not an identical parametrization; in particular, they do not share the same prior distribution. Precisely for this reason the agreement of the results documented in Section 4 constitutes significant empirical evidence: the fact that the extracted signal and the estimated power-law parameters coincide *despite* the differences in engine, kernel, regularization, and resolution indicates that, for the Campi Flegrei GNSS data analyzed here, the information contained in the series is sufficient to make the estimate substantially independent of these implementation choices. In particular, it is sufficient to override the pull of the GP priors.

4 Results

The results are organized in three parts, corresponding to as many levels of verification of the equivalence between the two baselines: the spectral validation, which compares the effect of the two methods on the spectrum of the residuals; the estimation of the noise parameters, which compares the quantities identifiable by the two estimators; and the agreement between the extracted deformation signals, which directly quantifies how much the two reconstructions of $f(t)$ coincide. Consistently with what was established in Section 3.4, the equivalence is verified on the extracted signal and on the noise parameters, not on the smoothness hyperparameters of the two kernels, which parametrize different covariance functions and are not directly comparable. All the results are reported for the station ACAE–LICO as a representative case and, in tabular form, for the entire network of seven stations.

4.1 Spectral validation

The first check concerns the effect of the two methods on the power spectral density (PSD) of the residuals. Figure 5 overlays, for the three components of ACAE–LICO, the PSD of the original residual and those of the residuals obtained after removing the deformation signal estimated by GP and by Kalman (residual $- f$), together with the power-law line of index κ estimated by Hector. The quantitative indicator is the spectral slope in the colored band, estimated by linear regression on a log–log scale. The two baselines preserve the spectral slope of the residual to the third decimal place, on all three components: GP and Kalman act on the spectrum in the same way. It must be specified that the PSD shown is that of the residual *net of the extracted signal* (residual $- f$): a method that correctly reconstructs the deformation removes power in the low-frequency band where the signal resides, leaving the noise at the higher frequencies unaltered. Consistently, the GP and Kalman curves (Figure 5) drop at the lowest frequencies, where each method has subtracted its own estimate of $f(t)$, and retrace the original residual along the power-law line in the colored band.

The two curves coincide over the whole high- and mid-frequency band, where neither of the two methods intervenes: the colored and white noise remains unaltered. The only differences, minimal at that, are located at the lowest frequencies,

Table 1: Spectral slope of the residuals of ACAE–LICO before filtering (original) and after filtering with GP and with Kalman, compared with the power-law index κ estimated by Hector.

Component	Original	After GP	After Kalman	κ (Hector)
East	−0.504	−0.503	−0.504	−0.536
North	−0.675	−0.673	−0.673	−0.778
Up	−0.757	−0.757	−0.757	−0.795

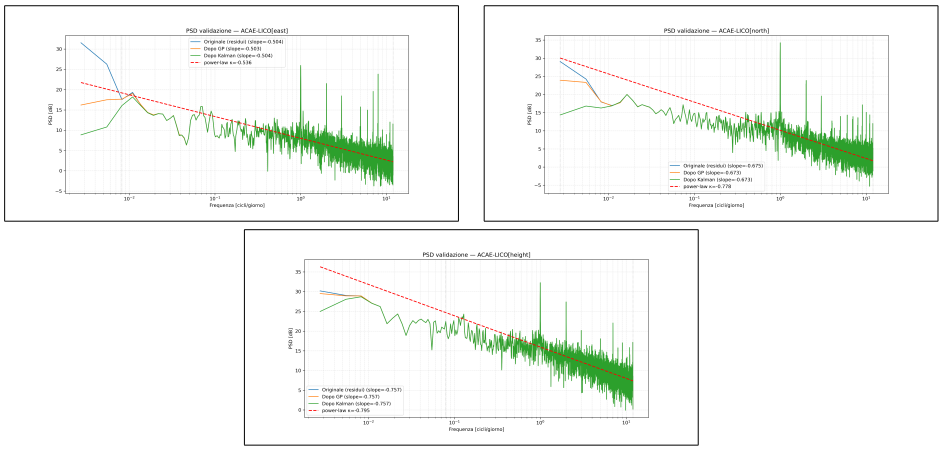


Fig. 5: Spectral validation for ACAE–LICO: PSD of the original residual and of the residuals after removal of the signal estimated by GP and by Kalman, with the power-law line of index κ (Hector). The slopes coincide between the two methods.

where each method has removed its own estimate of the deformation signal $f(t)$; their magnitude is consistent with the sub-millimeter agreement between the two signals documented in paragraph 4.3. Note that the empirical slope does not coincide with the Hector index κ, the more so the steeper κ is (compare East and Up): it is the spectral manifestation of the overestimation of the power-law by Hector discussed in Section 2.7, which both estimators correct by reducing the power-law scale factor, as quantified in the following paragraph.

4.2 Estimated noise parameters

Table 2 reports, for all twenty-one series, the power-law scale factor and the white-noise amplitude estimated by the two methods, with the Hector index κ alongside. The power-law scale factor (`pl_scale`) is the quantity on which the two methods are directly compared, and it is the central quantitative result. In both estimators it collapses to values between about 0.03 and 0.12, that is, about an order of magnitude below the unit value that would correspond to fully accepting the Hector power-law amplitude. The two methods agree not only in the order of magnitude but also in the *pattern*: the scale factor is systematically higher on the horizontal East components (around 0.09–0.12) and lower on the

North and Up components associated with the steepest indices κ (down to 0.03–0.05). In other words, GP and Kalman, starting from independent formulations, agree in diagnosing that Hector overestimated the power-law, and agree on *how much* and *where* the overestimation is greater. This agreement is all the more significant for the GP, whose scale factor is subject to a LogNormal prior centered on unity (Section 3.2): despite the prior pushing it toward the Hector value and the lower constraint letting it drop down to 0.01, the estimated scale factor settles stably around 0.05–0.12, far from both the prior and the constraint. The result is therefore data-driven and not imposed by the regularization, and it is precisely for this reason that it coincides with that of the Kalman, which has no prior. On the white noise the two methods behave differently, and the difference must be read with care. Both reject the collapse to zero produced by Hector (amplitude of the order of 10^{-4} mm, physically implausible) and bring the white noise back to a millimeter level. However, while the GP estimates it freely, obtaining values between 0.7 and 3.0 mm that vary from series to series, in the Kalman the white noise is not separately identifiable from the high-frequency tail of the power-law: the two components explain the same spectral band and the likelihood is insensitive to their partition. Consequently the white noise of the Kalman remains anchored to its own lower limit (0.548 mm, uniform across all the series), where the floor plays the role of regularizer that prevents its collapse, exactly as foreseen in Section 2.7. The millimeter value estimated freely by the GP therefore represents the informative estimate of the white noise, while that of the Kalman constitutes an imposed lower limit. An analogous behavior is observed on the Kalman lengthscale, which in eighteen series out of twenty-one remains close to 105 days, a value that coincides with the midpoint of the admitted interval and with the initialization: the smoothness scale of the signal too is weakly identified in the state-space formulation. This observation does not affect the signal extraction, as the following paragraph shows, and confirms that the comparison between the methods must be carried out on the reconstructed signal and not on the individual hyperparameters.

4.3 Agreement between the extracted deformation signals

The decisive check is the direct comparison between the two extracted deformation signals, $f_{\mathrm{GP}}(t)$ and $f_{\mathrm{K}}(t)$. Figure 6 overlays them for the three components of ACAE–LICO, together with the original series and the uncertainty bands at $\pm 2\sigma$. On the East and Up components the two curves are indistinguishable to the eye: both reconstruct the slow deformation trend, a signal of the order of hundreds of millimeters over four years, consistent with the bradyseismic dynamics, leaving out the dispersion of the noise (gray band), with very tight uncertainty bands. The agreement is quantified in Table 3 by the metrics computed between the two signals over all the common epochs: the RMS of the difference, the RMS of the signal, their ratio and the Pearson correlation. The RMS difference between the two signals is sub-millimeter or slightly higher in all the series (from 0.17 to 1.88 mm), against deformation signals that reach hundreds of millimeters; the Pearson correlation is greater than or equal to 0.999 in nineteen series out

Table 2: Parameters estimated by GP and Kalman for the twenty-one series: lengthscale of the signal kernel (ℓ, days), power-law scale factor (`pl_scale`) and white-noise amplitude (σ_w, standard deviation in mm). κ is the Hector power-law index.

			ℓ [d]		`pl_scale`		σ_w [mm]	
Station	Comp.	κ	GP	Kalman	GP	Kalman	GP	Kalman
ACAE–LICO	East	−0.536	64.7	105.3	0.104	0.095	1.018	0.548
ACAE–LICO	North	−0.778	139.1	105.1	0.059	0.039	1.510	0.548
ACAE–LICO	Up	−0.795	91.7	105.1	0.056	0.037	2.965	0.548
IPPO–LICO	East	−0.542	88.6	105.1	0.100	0.094	0.994	0.548
IPPO–LICO	North	−0.549	89.1	92.3	0.101	0.091	1.063	0.548
IPPO–LICO	Up	−0.766	98.5	105.1	0.057	0.041	3.003	0.548
PIS1–LICO	East	−0.525	86.8	44.3	0.103	0.100	0.925	0.548
PIS1–LICO	North	−0.625	131.6	105.0	0.079	0.067	1.064	0.548
PIS1–LICO	Up	−0.746	106.9	105.0	0.059	0.043	2.911	0.548
STRZ–LICO	East	−0.503	100.0	105.1	0.120	0.109	0.735	0.548
STRZ–LICO	North	−0.799	84.8	105.0	0.054	0.036	0.974	0.548
STRZ–LICO	Up	−0.756	119.5	105.1	0.057	0.042	2.072	0.548
ARFE–LICO	East	−0.482	44.4	105.2	0.124	0.118	0.717	0.548
ARFE–LICO	North	−0.902	126.3	105.0	0.048	0.027	0.995	0.548
ARFE–LICO	Up	−0.717	93.7	105.1	0.062	0.048	1.750	0.548
RITE–LICO	East	−0.496	100.7	105.1	0.113	0.112	0.845	0.548
RITE–LICO	North	−0.818	112.0	105.0	0.054	0.035	1.218	0.548
RITE–LICO	Up	−0.723	108.8	105.1	0.063	0.047	2.931	0.548
VICA–LICO	East	−0.551	88.4	106.1	0.105	0.090	0.736	0.548
VICA–LICO	North	−0.524	40.6	49.8	0.109	0.099	0.766	0.548
VICA–LICO	Up	−0.742	93.2	105.0	0.060	0.044	1.737	0.548

of twenty-one. The ratio between the RMS of the difference and the RMS of the signal remains below 2% in all the components that exhibit appreciable deformation. In physical terms: the two methods differ on a sub-millimeter scale, while the bradyseismic signal they must extract is of tens-of-centimeters order; the discrepancy lives entirely within the noise and is geophysically irrelevant. Four series present a higher ratio (STRZ–LICO East 2.1%, RITE–LICO East 3.4%, VICA–LICO East 12.2%, ACAE–LICO North 15.6%). In all these cases the high value does not reflect a disagreement between the methods but the smallness of the extracted signal: the ratio has the RMS of the signal at the denominator, which for these components is small (down to 1.4 mm for VICA–LICO East), so that even a sub-millimeter absolute difference produces an apparently large ratio. The RMS of the difference in absolute value indeed remains contained (0.17–0.78 mm) even for these series. ACAE–LICO North, which presents both the highest ratio and the only correlation appreciably lower than unity (0.988), constitutes the most marked case: it is a component with a weak trend and a modest deformation signal, in which the two reconstructions separate slightly at the local peaks, as visible in the central panel of Figure 6. Even in this case, however, the absolute difference is below the millimeter and does not alter the

Table 3: Agreement between the signals extracted by GP and Kalman for the twenty-one series: RMS of the difference, RMS of the signal, their ratio and Pearson correlation.

Station	Comp.	RMS diff. [mm]	RMS signal [mm]	Ratio [%]	Pearson r
ACAE–LICO	East	0.605	86.67	0.70	0.99998
ACAE–LICO	North	0.779	5.00	15.60	0.98783
ACAE–LICO	Up	1.268	108.58	1.17	0.99995
IPPO–LICO	East	0.456	56.83	0.80	0.99997
IPPO–LICO	North	0.276	24.27	1.14	0.99994
IPPO–LICO	Up	0.392	31.39	1.25	0.99993
PIS1–LICO	East	1.071	73.81	1.45	0.99990
PIS1–LICO	North	0.454	48.85	0.93	0.99996
PIS1–LICO	Up	1.093	111.87	0.98	0.99995
STRZ–LICO	East	0.574	27.99	2.05	0.99979
STRZ–LICO	North	0.742	88.59	0.84	0.99997
STRZ–LICO	Up	0.869	113.64	0.76	0.99997
ARFE–LICO	East	0.491	63.32	0.78	0.99997
ARFE–LICO	North	0.673	41.31	1.63	0.99987
ARFE–LICO	Up	0.841	83.75	1.00	0.99995
RITE–LICO	East	0.396	11.57	3.43	0.99941
RITE–LICO	North	0.404	47.68	0.85	0.99997
RITE–LICO	Up	1.883	194.39	0.97	0.99996
VICA–LICO	East	0.173	1.42	12.18	0.99258
VICA–LICO	North	0.647	52.81	1.23	0.99993
VICA–LICO	Up	0.273	32.97	0.83	0.99999

reading of the signal. Overall, the three checks agree: the two methods modify the spectrum in the same way, estimate the same power-law correction factor and extract the same deformation signal to within a fraction of a millimeter. This despite their differing in inference engine, signal kernel, regularization, temporal resolution and treatment of the white noise. The robustness of the extracted signal with respect to all these choices is the central empirical conclusion of this work.

5 Discussion

The central empirical result of this work is the robustness of the extracted deformation signal with respect to the inference engine. The three checks of Section 4 converge: for the same generative model, the same noise parameters inherited from Hector, and the same admissible support of the parameters, a batch Gaussian Process and a recursive Kalman filter act on the spectrum in the same way, estimate the same power-law correction factor, and extract the same deformation signal to within a fraction of a millimeter. This happens despite the two baselines differing in inference engine, signal kernel, regularization, temporal resolution, and treatment of the white noise, and despite their not sharing the same prior distribution. For the Campi Flegrei GNSS data analyzed here, therefore,

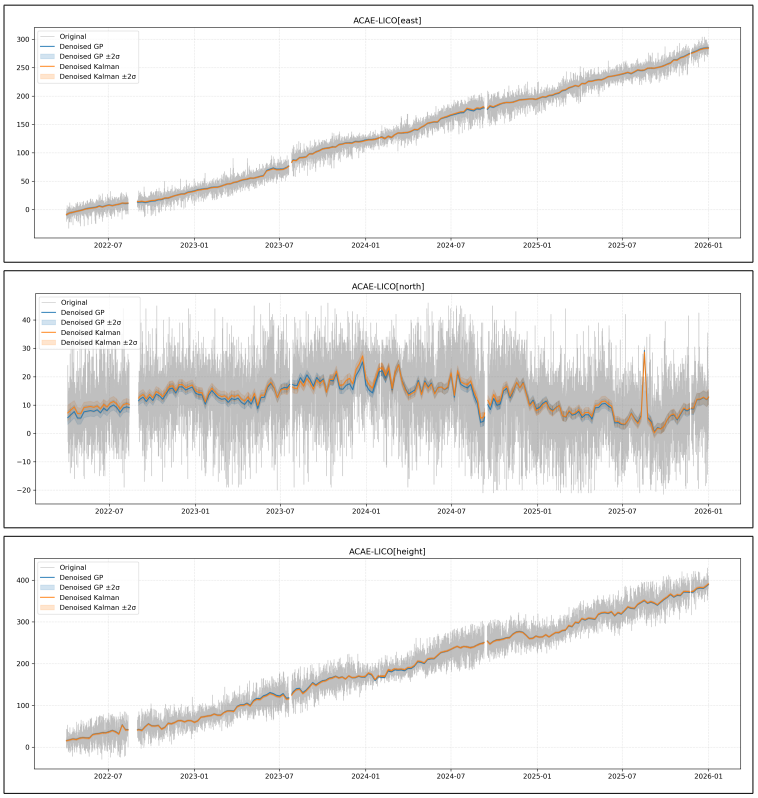


Fig. 6: Deformation signal extracted by GP (blue) and Kalman (orange) for the three components of ACAE–LICO (east, north and height), with the original series (gray) and uncertainty bands at $\pm 2\sigma$.

the information contained in the series is sufficient to make the estimate of $f(t)$ substantially independent of these implementation choices.

This independence has a direct consequence for the formulation of the problem. An approach in which one method guides the other, for example by initializing or constraining the Kalman filter with the Gaussian Process estimate, would make sense only if the two estimators contained complementary information about the signal. Since instead the two baselines converge on the same reconstruction of $f(t)$ and on the same power-law scale factor, there is no information that one could provide to the other to improve the estimate: the guided approach is redundant for this task. Note that this conclusion is confined to the data and the model considered here and does not constitute a general claim about the two methods; it is the specific configuration of the problem, namely a slow signal well separated in frequency from the noise and series that are sufficiently informative, that makes them equivalent.

6 Conclusions and future developments

We have presented a preprocessing chain for hourly GNSS time series of Campi Flegrei and two inference baselines, a Gaussian Process in batch formulation and a state-space Kalman filter, for extracting the residual slow deformation signal.

Several developments follow from this result. On the *preprocessing* side, the directions already indicated in Section 2.7 apply. On the *signal modeling* side, two directions appear promising. The first is symbolic regression [11], which would allow the search for an analytical and interpretable functional form of the extracted deformation signal, replacing the non-parametric description with an explicit expression. The second, viable when a physical model of the source is available in the form of a differential equation, is the use of Physics-Informed Neural Networks [24], which would constrain the signal estimate to comply with the physical laws governing the deformation.

A third direction concerns uncertainty quantification and signal reconstruction through generative models. Diffusion models (Denoising Diffusion Probabilistic Models) [17] and flow matching [22], which learn to transform noise into samples from the data distribution, offer a framework for generating ensembles of plausible signal reconstructions and quantifying their uncertainty in a distributional way, complementary to the posterior bands provided by GP and Kalman.

Finally, the extracted signal constitutes the starting point for a series of downstream analyses: estimation of the deformation velocity and of the true slope of the signal, imputation of the missing epochs, and forecasting of the evolution of the bradyseism [8], possibly accompanied by explainability techniques (XAI) [32] to identify the factors driving the predictions. These applications, enabled by the availability of a robust deformation signal cleaned of noise, constitute the natural continuation of the present work.

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